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a

In Computer Organization and Design, which of the following modifications must be made in a computer to increase its processing speed?

Options:

1. Connecting an external hard drive to the computer
2. Increasing the size of the random access memory
3. Upgrading to the solid-state drive

☐ 1 and 3

☐ 1 and 2

☒ 2 and 3



☐ All of these

Reset Answer



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In Computer Organization and Design, if you are required to perform 1 million reads and 1 million write operations in the minimum possible time, then which of the data storages will you not use to perform this task:

1. Registers
2. RAM
3. Hard disk
4. Floppy disk

☐ 1

☐ 2

☐ 3

☐ 4



In Computer Organization and Design, which of these approaches of performing interprocess communication is depicted by the following descriptions:

1. It is a unidirectional data channel that is implemented through a file on the file system.
2. It is a structure that synchronizes multiple processes that are acting on shared resources.
3. It is created by a virtual disk in RAM and allows processes to communicate with each other.
4. It requires the formatting of data that is written through it for preserving message boundaries.

☐ 1. Named pipes 2. Semaphores 3. Shared memory 4. Socket



☐ 1. Socket 2. Semaphores 3. Shared memory 4. Named pipes

☐ 1. Named pipes 2. Semaphores 3. Sockets 4. Shared memory

☐ None of these

Question 4

Max. score: 4.00

Mike is developing firmware for a device that requires setting specific bits within a configuration register to enable or disable certain features. He needs to use bitwise operators and shift operators in Embedded C to achieve the desired bit manipulations. He has a configuration register `configReg` with 8 bits, and he needs to set the 2nd and 5th bits to 1 while keeping the other bits unchanged. Which of the following code snippets accomplishes this task correctly?

☐ `configReg |= (1 << 2) | (1 >> 5);`☐ `configReg &= ~((1 << 2) | (1 << 5));`☒ `configReg |= (1 << 2) | (1 << 5);` ✓☐ `configReg &= ((1 << 2) | (1 << 5));` Reset Answer

In an operating system, consider the following scenario:

Scenario:

There are six resources. 'p' number of processes are competing for these resources. Each process may require three resources.

What is the maximum value of 'p' such that no deadlock occurs at all?

☐ 4

☐ 3

☒ 2

☐ 1

Question 6

Max. score: 4.00

John is working on a project that requires manipulating individual bits within variables to control various hardware peripherals of a microcontroller. He needs to use bitwise operators and shift operators in Embedded C to achieve the desired bit manipulations. He wants to set the 3rd bit (counting from the right) of a variable `value` to 1 while keeping the other bits unchanged. Which of the following code snippets accomplishes this task correctly?

☒ `value |= (1 << 3);` ☐ `value &= ~(1 << 3);`☐ `value ^= (1 << 3);`☐ `value &= (1 << 3);` Reset Answer



Question 7

Max. score: 2.00

A, B, C, D, E, F, G, and H are sitting in a round circle. A is sitting at a position second to the right of E who is the neighbor of C and G. There are 2 people in between A and G and 3 people in between A and F. If A and D are not neighbors and H is between B and D then who is to the right of F?

☐ A

☒ G

☐ D

☐ C

Reset Answer

Question 8

Max. score: 4.00

In data structures, if the input stack is $[1, 2, 3, 4, 5]$ and $n=5$, then what is the output of the following code?

Code:

```
void hacker(Stack S, int n,int c=0)
{
    if (S is empty || c == n)
        return;
    character x = S.top();
    S.pop();
    hacker(S, n, c+1);
    if (c != n/2)
        S.push(x);
}
```




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12

```
S.pop();  
hacker(S, n, c+1);  
if (c != n/2)  
    S.push(x);  
}
```

☐ 1 3 4 5

☒ 1 2 4 5

☐ 1 2 3 4

☐ 2 3 4 5

Reset Answer



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```
class Node {  
  
    // constructor...  
    Node(value) {  
        data = value  
        next = NULL  
    }  
}  
  
FUNCTION print_rev( h )  
    IF h==NULL  
        RETURN  
    ENDIF  
    // Line 12  
    // Line 13  
ENDFUN
```

Question 9

Max. score: 4.00

In Data Structures Ben is working on Single Linked List. The below pseudocode contains a function print_reverse that takes the head node(h) of a single linked list as input and prints the data of the nodes in reverse order.

Note: The data print_rev function prints from the data of the last node to the data of the first(head) node.

Example

Consider a Single Linked List 1 -> 2 -> 3 -> 4

The function print_reverse prints as 4, 3, 2, 1

Pseudocode

```
class Node {  
  
    // constructor...  
    Node(value) {  
        data = value  
        next = NULL  
    }  
}
```

Note: In the above pseudocode the class Node represents the structure of each node in the single linked list.

But the pseudocode was incomplete. If he is asked to complete the pseudocode then which one of the following code snippets will best fit in Line 12 and Line 13 such that the function print_rev prints the data of the nodes in the reverse order?

Code snippets

1.

```
print_rev( h.next )
PRINT h.data
```

2.

```
PRINT h.data
print_rev( h.next )
```

3.

```
print_rev( h)
```



4.

```
PRINT h.data  
print_rev( h)
```

☐ 1



☐ 2

☐ 3

☐ 4

Reset Answer

In Digital Computer Organization and Design, what is the value of variable *a* if three threads are reading the *a++* operation in the following code:

Code:

```
int a = 9;  
a++;
```

☐ 10 . . .

☐ 11

☐ 9 and 8

☐ 10 and 11 . . .



```
#include <stdio.h>
```

```
int main() {
```

```
    unsigned int data = 0xA5C349;
```

```
    int patternDetected = 0;
```

```
    if ((data & 0x0F) == 0x0A) {
```

```
        printf("Pattern detected: 0x0A\n");
```

```
        patternDetected = 1;
```

```
    }
```

```
    if ((data & 0xFF) == 0xC3) {
```

```
        printf("Pattern detected: 0xC3\n");
```

```
        patternDetected = 1;
```

```
    }
```

```
    if (!patternDetected) {
```

```
        printf("No pattern detected\n");
```

```
    }
```



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```
printf("No pattern detected\n");  
}  
  
return 0;  
}
```

☐ Pattern detected: 0A

☐ Pattern detected: C3

☒ No pattern detected



☐ Error

Reset Answer

Code

```
#include <stdio.h>
#include <string.h>
void hack_func(char* str, int n){
    int size1 = strlen(str);
    int i;
    int size2;
    if (size1 % n != 0) {
        printf("Wrong Input");
        return;
    }
    size2 = size1 / n;
    for (i = 0; i < size1; i++) {
        if (i % size2 == 0)
            printf(" ");
        printf("%c", str[i]);
    }
}
```



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```
int main(){  
    char* s = "Welcometohackerearth";  
    hack_func(s, 5);  
    getchar();  
    return 0;  
}
```

☐ Welcometohackerearth

☒ Welc omet ohac kere arth



☐ Welco metoh acker earth

☐ Wrong Input

Question 13

Max. score: 2.00 Negative score: 2.00

Suppose that Mike got a question based on an array. Which of the following option correctly states, "An array of 50 pointers to functions that take an integer argument and return an integer"?

☒ `int (*a[50])(int);`

☐ `int (**a[50])(int);`

☐ `int (a*[50])(int);`

☐ None of the above

Reset Answer

Question 14

Max. score: 50.00

Hiring drive

You are looking to hire front-end and back-end developers for a project that requires exactly N front-end developers and exactly M backend developers. You receive applications from $(N+M)$ candidates.

Conditions

- Every developer can work on either the front end or back end but not on both.
- The cost of each developer can be different. The cost of the i^{th} developer is as follows:
 - Front-end: F_i dollars.
 - Back-end: B_i dollars.

[< Previous Question](#)

[Next Question >](#)

C (gcc 10.3)

Auto-complete ready!

Autosaved

```
1 > #include<stdio.h>...
4 int solve (int N, int M, int* F, int* B) {
5     // write your code here
6 > }
```

Ln 1, Col 1, Visual Studio

[View test cases ^](#)

Run and save

▼ hiring frontend and backend with min cost (n+m)

```
int solve(vector<int> &f,vector<int> &b,int n,int m){
    vector<pair<int,int>> temp;
    for(int i=0;i<n+m;i++){
        temp.push_back({f[i],b[i]});
    }
    int ans=0;
    {
        sort(temp.begin(),temp.end(),[&](const pair<int,int> &a,
            return a.first-a.second < b.first-b.second;
        ));
        for(int i=0;i<n+m;i++){
            if(i<n){
                ans+=temp[i].first;
            }
            else{
                ans+=temp[i].second;
            }
        }
        return ans;
    }
}
```

Digits sum

Given a number N in string format and another number K . Calculate the sum of the digits in N where the index of each digit (with the indexing starting from 0 from left) is a multiple of K . Continue to apply this function recursively until the sum is reduced to a single digit.

Function description

Complete the `solve()` function. The function takes 2 parameters and returns an integer denoting the answer to the question:

- N : Represents the number in string form
- K : Represents an Integer.

Input format for custom testing

C (gcc 10.3)

✓ Auto-complete ready!

⬆ Autosaved ↺ ⚙

```
1 > #include<stdio.h>...
4 int solve (char* N, int K) {
5     // Write your code here
6 > }
```

Ln 1, Col 1, Visual Studio

[< Previous Question](#)

[Next Question >](#)

[View test cases ^](#)

Run and save

Sample input

```
2
123456789 3
123456789 7
```

Sample output

```
1
9
```

Explanation

The first line contains the total number of test cases, $T = 2$.

Test case 1

Given

- $N = 123456789$

C (gcc 10.3)

✓ Auto-complete ready!

⬆ Autosaved ↺ ⚙

```
1 > #include<stdio.h>...
4 int solve (char* N, int K) {
5     // Write your code here
6 > }
```

Ln 1, Col 1, Visual Studio

[< Previous Question](#)

[Next Question >](#)

[View test cases ^](#)

[Run and save](#)

▼ DigitSum till one digit left

```
int solve(string &str,int &k){  
    int sum =0;  
    for(int i=0;i<str.length();i+=k){  
        sum+=str[i] - '0';  
    }  
    if(sum<10) return sum;  
    string s = to_string(sum);  
    return solve(s,k);  
}
```


GDP of the countries

You are given a table *countries_metrics* that holds details like the economical metrics of various countries.

Task

Write an SQL query to fetch the names and the corresponding GDP values of the countries that have an average GDP value greater than or equal to 20 (20 Trillion US dollars). Round the GDP values off to their nearest integer values.

Arrange the output in decreasing order of the GDP value, and then in alphabetical order of the country's name.

Notes

- Strictly saying that the numeric values given in the sample input table are in Trillions of US

New Submission

All Submissions

Save

MySQL 8.0



1

Ln 1, Col 1, Visual Studio

Execute & Test

Submit code

- Strictly saying that the numeric values given in the sample input table are in Trillions of US Dollars. For example, 10 implies 10 Trillion US dollars.
- Formula to calculate the GDP; $GDP = C + G + I + NX$; where **C** stands for consumption or all private consumer spending within a country's economy,
- **G** stands for total government expenditures, including salaries of government employees, road construction/repair, public schools, and military expenditures,
- **I** stand for the sum of a country's investments spent on capital equipment, inventories, and housing,
- **NX** stands for net exports or a country's total exports less total imports.

New Submission

All Submissions

Save

MySQL 8.0



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Ln 1, Col 2, Visual Studio

Execute & Test

Submit code

Table description

Input format

Table: *countries_metrics*

Field	Type	Description
country_id	Integer	Represents the unique ID of the country
country_name	String	Represents the unique name of the country
consumption	Integer	Represents the consumption of goods and services within a country's economy
government_expenditures	Integer	Represents the total amount of money spent by the government
country_investment	Integer	Represents the sum of investments made by the country
country_netexports	Integer	Represents the net exports of the country

Output format

Field	Type	Description
country_name	String	Represents the unique ID of the country

New Submission

All Submissions

Save

MySQL 8.0



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Ln 1, Col 3, Visual Studio

Execute & Test

Submit code

Output format

Field	Type	Description
country_name	String	Represents the unique ID of the
GDP	Integer	Represents the country's gross domestic product value

Example

Input table

Table: *countries_metrics*

country_id	country_name	consumption	government_spending
101	India	10	2
102	France	8	3
103	China	16	6
104	Japan	9	2
105	US	11	8

New Submission

All Submissions

Save

MySQL 8.0



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Ln 1, Col 4, Visual Studio

Execute & Test

Submit code

country_id	country_name	consumption	government_
101	India	10	
102	France	8	
103	China	16	
104	Japan	9	
105	US	11	
103	India	10	
101	China	8	

Output table

country_name
US
China
India

New Submission

All Submissions

Save

MySQL 8.0



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Ln 1, Col 5, Visual Studio

Execute & Test

Submit code

```
SELECT country_name,  
ROUND(AVG(consumption+government_expenditures+country_investment+country_netexports)) as GDP  
FROM countries_metrics  
GROUP BY country_name  
HAVING GDP >= 20  
ORDER BY GDP desc, country_name;
```